



Analysis of Competing Species Model

Abubakar Othman (F221611)

Mathematical Modelling I (25MAP111)

Department of Mathematical Sciences

Abstract

The report investigates the dynamics of two competing species undergoing harvesting, using a model based on the Lotka-Volterra competitive species model and the concept of Maximum Sustainable Yield. Using a value for MSY adapted to the competitive system, I was able to carry out the analysis. The model is non-dimensional, allowing for simpler analysis of relative parameters.

Results show that increasing competition or harvest can lead to different population outcomes, including coexistence, the extinction of one species, or the extinction of both. Harvesting at 80% of the MSY provided the most optimal yield as it was a safe approach, as well as the most financially beneficial outcome.

The findings showed the importance of each parameter in the system and how they affect the system as a whole, as well as the consequences of aggressive fishing. Our findings also showed that we have to consider how much one species affects another to determine how much we will harvest.

Contents

1	Introduction	3
2	Background	3
2.1	Single Species Logistic Growth Model	3
2.2	Two Species competition Lotka-Volterra Model	4
2.3	Harvesting and Management	6
3	Model Description	6
3.1	Model Assumptions	6
3.2	Dimensional Model	7
3.3	Maximum Sustainable Yield and Harvest Rates	7
3.4	Non-Dimensional Model	8
4	Results and Discussion	9
4.1	Equilibrium Points	9
4.1.1	Stable Coexistence	10
4.1.2	Competitive exclusion	10
4.1.3	Extinction	13
4.1.4	Comparison of Equilibrium	13
4.2	Influence of Harvesting	13
4.2.1	Maximum Sustainable Yield (MSY)	14
4.2.2	Effect of Different Harvest	14
4.3	Parameter Dependence	16
4.3.1	Competition Coefficients	16
4.3.2	Growth Rate Ratio	17
4.3.3	Harvest Amount	17
5	Conclusion	18
A	MSY Derivation	19
A.1	Single MSY Derivation	19
A.2	Joint MSY Derivation	19
B	Acknowledgements	20
	References	21

1 Introduction

Ecological systems often contain multiple species that rely on similar resources to survive. With limited resources, the growth of one species naturally affects the development of another, leading to competition. Understanding how such species interact with each other is essential for predicting a population's long-term growth; this is useful for managing fish harvests and ensuring we sustain a healthy population that can sustainably manage the ecosystem.

We can leverage mathematical models to provide a systematic way to describe and analyse these interactions, allowing us to predict how the population will grow. In this paper, a model is developed for two fish species inhabiting the same lake and competing for limited resources.

The model is based on the Lotka-Volterra equations, a simple model of the population dynamics of species competing for a common resource. This model will also help us determine the conditions under which both species can coexist, making it a valuable tool for designing sustainable harvesting policies.

The report focuses on a lake that supports two fish species competing for the same food resource. Fishermen will remove a proportion of each species annually; our goal is to develop a model of the two fish's growth and advise fishermen on how much they can harvest without driving the species to extinction.

2 Background

2.1 Single Species Logistic Growth Model

For the competition equations, the logistic equation is the basis; it is used by ecologists in the form

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) \quad (1)$$

where:

- $\frac{dN}{dt}$ is the rate of change in the size of the population N , at a given moment in time, t
- N is the current population size
- t is the given moment in time
- r is the growth rate
- K is the carrying capacity; the number of individuals a population can support indefinitely

In this equation, if K is much larger than N , the population can grow quickly, but if N is greater than K , the population will shrink. If N and K are equal, the population will remain constant over time.

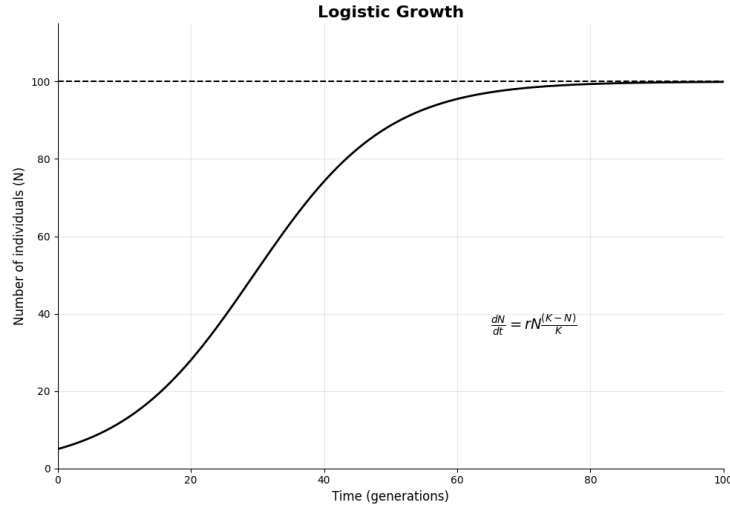


Figure 1: Logistic Growth Curve

The logistic model provides an asymptotic approach to population size, represented by K ; however, in nature, a variety of factors, such as extreme weather and resource availability, cause greater fluctuations in populations. We can see this from [1]

2.2 Two Species competition Lotka-Volterra Model

The Lotka-Volterra competition model builds on the logistic model of a single population. It begins with separate logistic models for the populations of the two competing species. The model describes how two species interact through resource limitation. The model is written as

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1} \right), \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2} \right),\end{aligned}\tag{2}$$

where:

- $\frac{dN_1}{dt}$ and $\frac{dN_2}{dt}$ are the rate of change in the populations N_1 and N_2 at a given time t
- N_1 and N_2 are the current population sizes of species one and species 2
- t is the given moment in time
- r_1 and r_2 are the growth rates of the populations
- K_1 and K_2 are the carrying capacities of each of the species

- α_{12} is the competition coefficient that describes the effect of species two on species 1
- α_{21} is the competition coefficient that describes the effect of species one on species 2

The competition coefficient tells us about the competition dynamics between the two species:

- If $\alpha_{12} < 1$, an individual of species 2 has a smaller effect on species one than another individual of species one would have.
- If $\alpha_{12} > 1$, an individual of species 2 has a larger effect on species one than another individual of species one would have.
- If $\alpha_{12} = 1$, an individual of species 2 has a similar effect on species one as another individual of species one would have.

In this study, the model represents two fish species sharing a food source in a lake. As each species' growth depends not only on its own density but also on the presence of the other species. The system is in equilibrium when both growth rates are zero. Depending on the parameter values, there are several possible outcomes:

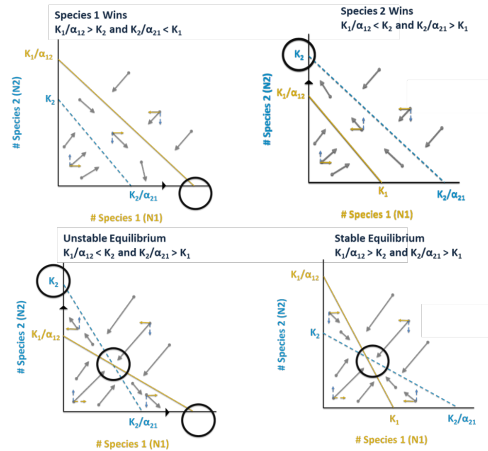


Figure 2: Logistic Growth Curve:

https://bio.libretexts.org/Courses/Gettysburg_College/01%3A_Ecology_for_All/15%3A_Competition/15.05%3A_Quantifying_Competition_Using_the_Lotka-Volterra_Model

- **Stable coexistence** - both species survive at equilibrium when $\alpha_{12} < \frac{K_1}{K_2}$ and $\alpha_{21} < \frac{K_2}{K_1}$

- **Competitive exclusion** - one species drives the other to extinction when $\alpha_{12} < \frac{K_1}{K_2}$ and $\alpha_{21} > \frac{K_2}{K_1}$, or vice versa
- **Dependence on initial conditions** - either species can win depending on who starts ahead when $\alpha_{12} > \frac{K_1}{K_2}$ and $\alpha_{21} > \frac{K_2}{K_1}$

We can see this from [2] and [3]

2.3 Harvesting and Management

Human activity alters population dynamics by removing individuals from the environment. Fishing can reduce population sizes and, if uncontrolled, can lead to a collapse. To allow fishing while keeping populations stable, we need to identify harvesting rates that maintain stable populations over time. In a single-species setting, we can modify the logistics model so that a certain number of individuals are continuously removed:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) - H \quad (3)$$

When H is constant, the population will be at equilibrium when the number removed equals the population growth rate. The equilibrium population size under a particular harvesting regime is found when the population is not growing, i.e., when the growth rate equals the harvest rate. The maximum sustainable yield (MSY) is the most significant yield that can be taken from a population at equilibrium. Harvesting at the MSY is dangerous, as you are harvesting at the knife's edge. The formula for the maximum sustained harvest H is:

$$H = \frac{Kr}{4} \quad (4)$$

The maximum economic yield (MEY) is the value of the largest positive difference between total revenue and total costs of fishing. MEY is typically achieved at catches that are 10-20% smaller than the MSY. We can see this from [4] and [5]

3 Model Description

3.1 Model Assumptions

The model considers a lake with two fish species. Every year, fishermen remove fish from the lake during the summer. The two species are also competing for the same food source. The following assumptions have been made:

1. Each species would follow logistic population growth if the other were absent
2. The weather is consistent throughout the year

3. Competition is due to species competing over the same resource
4. The species do not have a predator-prey relationship
5. The effect on which species i affects species j , where $i \neq j$, is constant, and the model doesn't account for any potential changes in this effect
6. Once a population reaches equilibrium, it will not grow indefinitely and stay at that stable population size

These assumptions simplify the system and allow analysis of how harvesting and competition can affect the population stability.

The model is based on the Lotka-Volterra Model and the idea of Maximum Sustainable Yield (MSY) and Maximum Economic Yield (MEY)

3.2 Dimensional Model

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1} \right) - H_1, \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2} \right) - H_2,\end{aligned}\tag{5}$$

Where:

- $\frac{dN_1}{dt}$ and $\frac{dN_2}{dt}$ are the rate of change in the populations N_1 and N_2 at a given time t
- N_1 and N_2 are the current population sizes of species one and species 2
- t is the given moment in time
- r_1 and r_2 are the growth rates of the populations
- K_1 and K_2 are the carrying capacities of each of the species
- α_{12} is the competition coefficient that describes the effect of species two on species 1
- α_{21} is the competition coefficient that describes the effect of species one on species 2
- H_1 and H_2 is the amount harvested from each population

3.3 Maximum Sustainable Yield and Harvest Rates

For this section, we will make use of the concept of Maximum Sustainable Yield (MSY) and Maximum Economic Yield (MEY) that we talked about in Section 2.3

To incorporate harvesting into the model, we will use the MSY to derive H_1 and H_2 in Equation 5. H_1 and H_2 are derived from Equation 4 so that they can be used in the context of the competing species. Harvest rates depend on

the effect one species has on another's growth. We derive a model for the MSY for a competing species as:

$$\begin{aligned} H_{MSY_1} &= \frac{r_1}{4K_1}(K_1 - \alpha_{12}N_2)^2, \\ H_{MSY_2} &= \frac{r_2}{4K_2}(K_2 - \alpha_{21}N_1)^2 \end{aligned} \quad (6)$$

To determine the amount we will harvest, we will set a limit so that we do not exceed MSY and do not risk the fish population collapsing. Still, we will also consider the MEY, which is achieved at 80-90% of the MSY, so that we can achieve the highest possible profits. For the model, I will set it at 80% as a safer level to keep the fish population stable. The system also preserves the single-species case when equating a rival population to 0.

$$\begin{aligned} H_1 &= H_{MEY_1} = 0.8H_{MSY_1} \\ H_2 &= H_{MEY_2} = 0.8H_{MSY_2} \end{aligned} \quad (7)$$

3.4 Non-Dimensional Model

To simplify analysis and reduce the number of parameters, we can introduce our dimensionless model with dimensionless parameters. We can do this by taking our dimensional model from Equation 5 and defining dimensionless parameters for it:

$$\begin{aligned} n_1 &= \frac{N_1}{K_1}, & n_2 &= \frac{N_2}{K_2}, & \tau &= r_1 t, \\ \beta_{12} &= \frac{\alpha_{12}K_2}{K_1}, & \beta_{21} &= \frac{\alpha_{21}K_1}{K_2}, & \rho &= \frac{r_2}{r_1}, \\ h_1 &= \frac{H_1}{r_1 K_1} = \frac{(1 - \beta_{12}y)^2}{5}, & h_2 &= \frac{H_2}{r_1 K_2} = \frac{\rho(1 - \beta_{21}x)^2}{5} \end{aligned} \quad (8)$$

This is only valid for $\beta_{12}y < 1$ or $\beta_{21}x < 1$, as if the conditions were not satisfied, it would imply a negative carrying capacity, which we cannot have.

With these dimensionless variables, we can now get our dimensionless model:

$$\begin{aligned} \frac{dx}{d\tau} &= x(1 - x - \beta_{12}y) - h_1 \\ \frac{dy}{d\tau} &= \rho y(1 - y - \beta_{21}x) - h_2 \end{aligned} \quad (9)$$

Where:

- n_1 - which is the population of species one scaled down by the carrying capacity of species 1

- n_2 - which is the population of species two scaled down by the carrying capacity of species 2
- τ - is the dimensionless time, scaled by the growth rate of species 1, r_1
- ρ - is the ratio of the growth rate of the two
- β_{12} - the dimensionless competition coefficient that describes the effect of species two on species one relative to the carrying capacities of both species
- β_{21} - the dimensionless competition coefficient that describes the effect of species one on species two relative to the carrying capacities of both species
- h_1 - the dimensionless harvest rates for species one scaled by the carrying capacity of species and the growth rate of species 1
- h_2 - the dimensionless harvest rates for species two scaled by the carrying capacity of species two and the growth rate of species 1

4 Results and Discussion

4.1 Equilibrium Points

The equilibrium states of the model occurs when both $\frac{dx}{d\tau}$ and $\frac{dy}{d\tau}$ from the dimensionless model in Equation 9 are:

$$\frac{dx}{d\tau} = \frac{dy}{d\tau} = 0$$

This gives us the equilibrium conditions:

$$\begin{aligned} x(1 - x - \beta_{12}y) &= h_1 \\ \rho y(1 - y - \beta_{21}x) &= h_2 \end{aligned}$$

Depending on the dimensionless parameters $(\beta_{12}, \beta_{21}, \rho, h_1, h_2)$ used, there are several outcomes for when the system is in equilibrium:

1. Stable coexistence
2. Competitive exclusion
3. Extinction

4.1.1 Stable Coexistence

Stable coexistence is when both species survive at equilibrium when $\beta_{12} < 1$, $\beta_{21} < 1$ and harvesting is done safely which is below the MSY

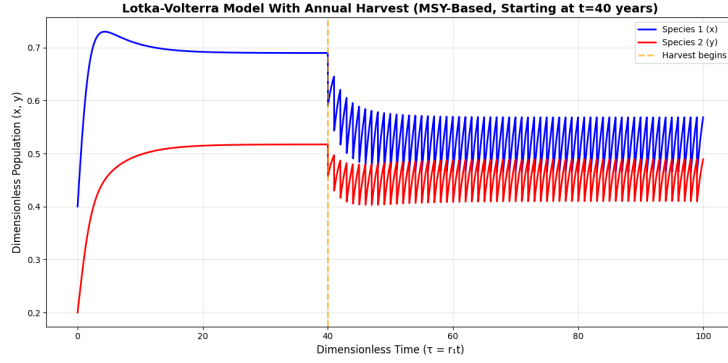


Figure 3: Time series showing stable coexistence between species

Figure 3 had model parameters:

$$\begin{aligned}(x_0, y_0) &= (0.4, 0.2), \\ (\beta_{12}, \beta_{21}, \rho, h_1, h_2) &= (0.6, 0.7, 1, 0.1549, 0.1037) \\ (h_{MSY_1}, h_{MSY_2}) &= (0.1936, 0.1296)\end{aligned}$$

As we can see since $\beta_{12}, \beta_{21} < 1$ and $h_i < h_{MSY_i}$ we achieve a stable coexistence. We can also see that, since we are harvesting safely at 80

4.1.2 Competitive exclusion

Competitive exclusion is when one species drives the other to extinction when $\beta_{12} > 1$ and $\beta_{21} < 1$, or vice versa. It can also be when one species is harvested too aggressively, such that the harvest is above the MSY

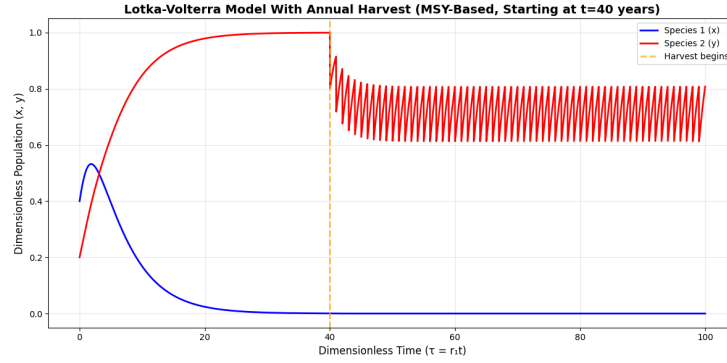


Figure 4: Time series showing competitive exclusion between species where species 2 dominates

Figure 4 had model parameters:

$$\begin{aligned}(x_0, y_0) &= (0.4, 0.2), \\ (\beta_{12}, \beta_{21}, \rho, h_1, h_2) &= (1.2, 0.7, 1, 0.1155, 0.1037) \\ (h_{MSY_1}, h_{MSY_2}) &= (0.1444, 0.1296)\end{aligned}$$

As we can see since $\beta_{12} > 1$, $\beta_{21} < 1$ and $h_i < h_{MSY_i}$ we achieve a competitive exclusion. We can also see that since we are harvesting safely at 80% of the MSY, the extinctions are due to the species' influence on each other. We can also see that since Population 2 is more dominant and eventually out of the picture, we can harvest more of species 2 in contrast to Figure 3

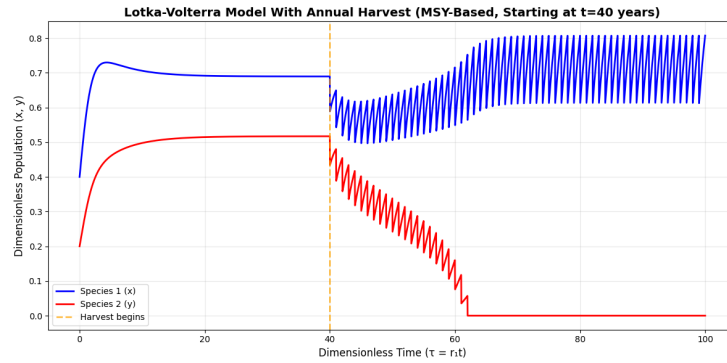


Figure 5: Time series showing competitive exclusion between species where species two is overharvested

Figure 5 had model parameters:

$$\begin{aligned}
(x_0, y_0) &= (0.4, 0.2), \\
(\beta_{12}, \beta_{21}, \rho, h_1, h_2) &= (0.6, 0.7, 1, 0.1549, 0.1426) \\
(h_{MSY_1}, h_{MSY_2}) &= (0.1936, 0.1296)
\end{aligned}$$

As we see since $\beta_{12}, \beta_{21} < 1$, $h_1 < h_{MSY_1}$ and $h_2 > h_{MSY_2}$. We are achieving competitive exclusion through over-harvesting. For population 2, we are harvesting above MSY, so it won't be able to recover. Because of this, population 2 is eventually out of the picture, allowing us to take more than we were taking before with population 1. We can also see that the change in only h_2 from Figure 3 caused such a big change in the graph seen in Section 4.1.1.

4.1.3 Extinction

extinction is when both species are driven to extinction, when they are both overharvested

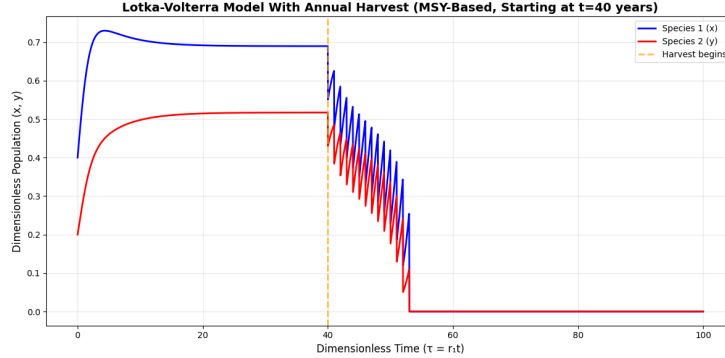


Figure 6: Time series showing extinction happens due to overharvesting

Figure 6 had model parameters:

$$\begin{aligned}(x_0, y_0) &= (0.4, 0.2), \\ (\beta_{12}, \beta_{21}, \rho, h_1, h_2) &= (0.6, 0.7, 1, 0.21296, 0.14256) \\ (h_{MSY_1}, h_{MSY_2}) &= (0.1444, 0.1296)\end{aligned}$$

As we can see since $\beta_{12}, \beta_{21} < 1$ and $h_i > h_{MSY_i}$ we achieve extinction. We can also see that, since we are harvesting aggressively at 110% of the MSY, extinctions are due to our overharvesting, which exceeds our MSY. We can see both populations decrease, and when they reach 0, we can no longer harvest.

4.1.4 Comparison of Equilibrium

The three equilibrium outcomes highlight how population persistence depends on both Competition strength and harvesting intensity.

When competition coefficients are moderate $\beta_{12}, \beta_{21} < 1$, harvesting is done safely $h_i < h_{MSY_i}$, and both species survive at a reduced but stable equilibrium population. Increasing competition or harvesting results in our other equilibrium outcome, so that either one species dominates or both are dominated.

4.2 Influence of Harvesting

In this section, we will see how harvesting is affected by its parameters and how it affects the system.

4.2.1 Maximum Sustainable Yield (MSY)

In a single-species system, the model achieves MSY when the population size equals half the species' carrying capacity, but in our two-species system, it works differently. From Section 3.3 and 3.4, we can see that our MSY for each species is:

$$\begin{aligned} h_{MSY_1} &= \frac{(1 - \beta_{12}y)^2}{4}, \\ h_{MSY_2} &= \frac{\rho(1 - \beta_{21}x)^2}{4} \end{aligned} \quad (10)$$

The expression shows that if there is stronger competition β_{ij} , then the MSY is reduced, as the adequate carrying capacity gets diminished. We can also see that if the population of the competitor species is significant, then the MSY is also reduced.

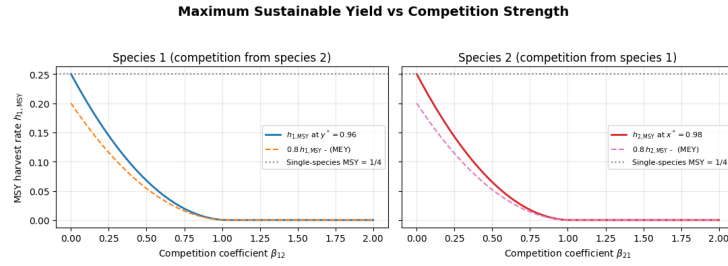


Figure 7: Visualisation of how MSY is affected by the Competition Coefficient

From Figure 7, we see that as the competition coefficient increases at a fixed population, MSY decreases. Both curves begin at 0.25, where the competition coefficient is 0. This is because, since there is no competition, we can treat it as a single-species MSY. We can also see that the MEY is directly related, as a similar shape is observed. We would also observe a similar trend if we fixed our competition coefficients and varied our populations, given the nature of $1 - \beta_{12}y$ and $1 - \beta_{21}x$.

We can also confirm that harvesting at approximately 80% of MSY is a good idea, as we can achieve similar yields with a much safer approach. As we harvest farther from the MSY, we are less likely to drive the species toward extinction. As well as harvesting more safely, we also harvested to maximise profits. Hence, the MEY is both a safer and profitable option than the MSY.

4.2.2 Effect of Different Harvest

When harvesting at a level that exceeds MSY, the removal rate exceeds the rate at which the population can grow, so the population can no longer maintain a positive equilibrium, this occurs when $h_i > h_{MSY_i}$, as this means we are

harvesting more than the calculated safe amount, which means a species cannot replenish its population fast enough, leading to Extinction or Competitive Exclusion.

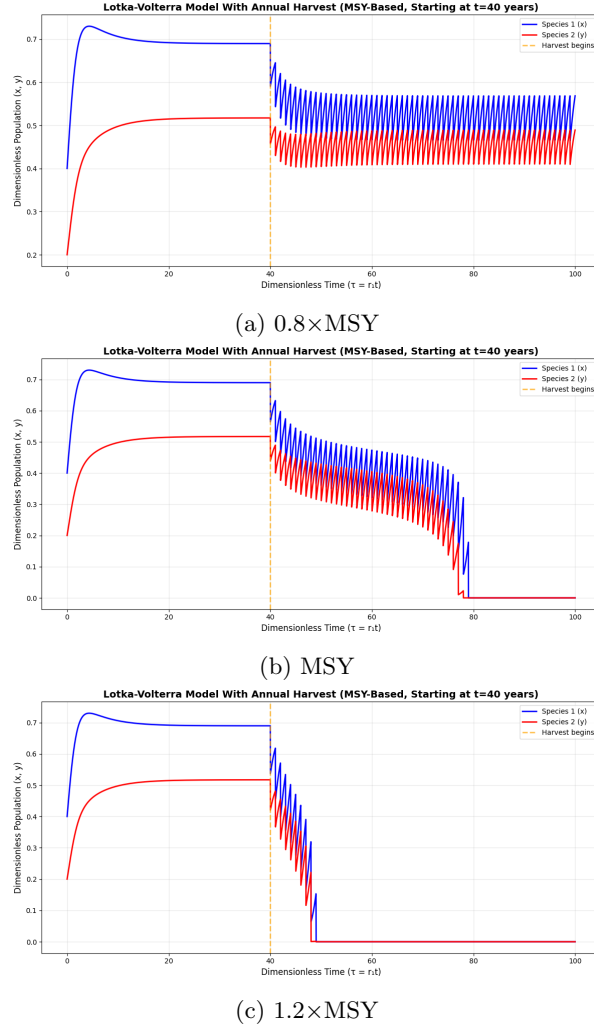


Figure 8: Graphs with varying harvests where other parameters are constant.

The graphs in Figure 8 show the time series trajectories for three harvesting levels: (a) $0.8 \times \text{MSY}$, (b) $1 \times \text{MSY}$, (c) $1.2 \times \text{MSY}$. Harvesting begins at $\tau = 40$ after equilibrium is reached.

- (a) At $0.8 \times \text{MSY}$, we have harvesting at our MEY, we can see that the populations settle to a stable equilibrium where they both coexist;
- (b) At $1 \times \text{MSY}$, we are harvesting at our MSY. We can see that our populations

are stabilising initially, but eventually the harvest is too high to sustain, as we do not have the population to support it. We can also see that our theoretical MSY overestimates the true safe yield

- (c) At $1.2 \times \text{MSY}$, we are harvesting over our MSY, causing both species to rapidly collapse, as we are harvesting more than the species can overcome

Table 1: Summary of population outcomes under different harvest regimes.

Harvest regime	Harvest rate h_i	Population outcome	Interpretation
$0.8 \times \text{MSY}$	$h_i < h_{\text{MSY}_i}$	Stable coexistence	Sustainable and profitable
$1.0 \times \text{MSY}$	$h_i = h_{\text{MSY}_i}$	Steady Extinction	Theoretical MSY but unstable in practice
$1.2 \times \text{MSY}$	$h_i > h_{\text{MSY}_i}$	Rapid extinction	Collapse of both species due to overharvesting

These results show that even harvesting at the MSY is not safe and overestimates the safe harvest. Operating below this level maintains stable coexistence, but above it we drive species into extinction. The results are considering the multiplier applied to both harvests, if it were used to one, we would get a Competitive Exclusion like in Figure 4

4.3 Parameter Dependence

In this section, we assess how each parameter affects equilibrium. This allows us to see which parameters have the most significant influence on the equilibrium outcomes.

4.3.1 Competition Coefficients

The competition determines the degree to which one species restricts the growth of the other. For smaller competition coefficients where β_{ij} , species self-regulate more strongly than they limit each other, resulting in a stable coexistence, with large equilibrium populations.

As the competition coefficient approaches 1, the restriction between species increases, which can cause a population to decrease significantly, whilst the other stays dominant, indicating a stable coexistence where one population is smaller or a competitive exclusion

When the competition coefficient of a population exceeds 1, coexistence is not possible, and the weaker species cannot survive, causing competitive exclusion even if harvest is minimal or absent.

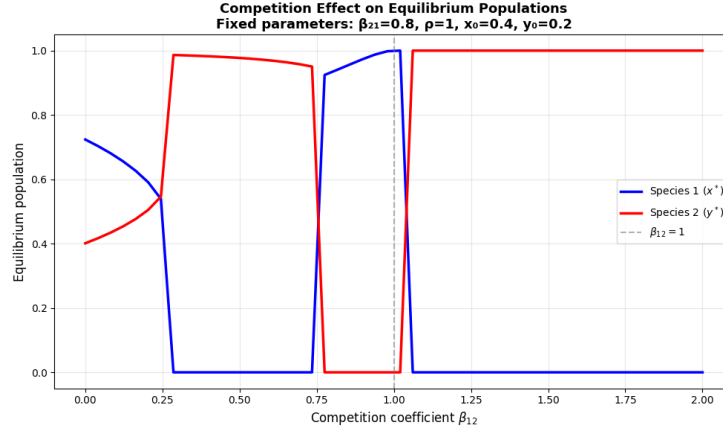


Figure 9: Visualisation showing how competition can affect a system

We can see that if we fix our system so that our values are constant whilst varying our competition coefficient β_{12} , we observe significant changes in the equilibrium. We can see that, depending on how much species 2 affects species 1, we experience different equilibria, indicating that the competition coefficient has a significant impact on the model. We can see that in some cases, a higher competition coefficient works to the advantage of species 1. This can also be due to the nature of the model, rather than how it would work in real life.

4.3.2 Growth Rate Ratio

The growth rate ratio $\rho = \frac{r_2}{r_1}$ scales the growth rates so that we can compare them more accurately. If the rate of growth of species 1 is significantly larger than that of species two, we will get higher values for species one and vice versa. If the values are equal, we have two species that progress at similar rates and can observe similar trends on the graphs; in this case, outcomes depend more on our competition coefficients.

4.3.3 Harvest Amount

The amounts we harvest, h_1, h_2 , are significant controllable parameters in the model that directly impact the equilibria, and are arguably the most critical parameters. We investigated the significance of this in Section 4.2. We can see that the system is sensitive to the amount harvested, as the value we take relative to the system's MSY determines the equilibrium type. We also noticed that harvesting at the MSY was unsafe, as the slight risk altered the kind of equilibrium the system reached.

5 Conclusion

This study developed and analysed a two species competition model with harvest based on the Lotka Volterra Competition and the Maximum Sustainable Yield. The results demonstrated that both competition and harvesting have substantial effects on the equilibrium outcomes. We could see that varying our competition coefficient and harvesting amount significantly affected the system's outcome.

The analysis showed that harvesting at the Maximum Economic Yield, approximately 80% of the MSY, achieved high yields, maintained a large population, and generated the most profit. This showed that harvesting at the MEY is the ideal option for fishermen, as it is both safe and most beneficial.

Overall, the competitive two-species model showed how different scenarios can play out in the competitive system and what the best strategy is for fishing in those scenarios.

A MSY Derivation

A.1 Single MSY Derivation

From [5] we can see that the MSY of a single species is achieved when the population size is half the carrying capacity, $N = \frac{K}{2}$. This is because if we take our logistic equation 1:

$$\begin{aligned}\frac{dN}{dt} &= rN \left(1 - \frac{N}{K}\right) - H \\ H(N) &= rN \left(1 - \frac{N}{K}\right) = rN - \frac{rN^2}{K}\end{aligned}$$

We can find the maxima of this with:

$$\begin{aligned}\frac{dH}{dN} &= r - \frac{2rN}{K} \\ N &= \frac{K}{2}\end{aligned}$$

We can then put this back into our H equation:

$$H\left(\frac{K}{2}\right) = \frac{rK}{4} \quad (11)$$

Thus, we have $\frac{rK}{4}$ as our MSY.

A.2 Joint MSY Derivation

For the two-species competition model with harvesting, we have dimensional equations from Equation 5:

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1}\right) - H_1, \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2}\right) - H_2,\end{aligned}$$

Following the same pattern as the previous section:

$$H_1(N_1) = r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1}\right) = r_1 N_1 - \frac{r_1 N_1^2}{K_1} - \frac{r_1 \alpha_{12} N_1 N_2}{K_1}$$

We can find out maxima with this:

$$\begin{aligned}\frac{dH_1}{dN_1} &= r_1 - \frac{2r_1 N_1}{K_1} - \frac{r_1 \alpha_{12} N_2}{K_1} = r_1 \left(1 - \frac{\alpha_{12} N_2}{K_1}\right) - \frac{2r_1 N_1}{K_1} \\ \frac{2N_1}{K_1} &= 1 - \frac{\alpha_{12} N_2}{K_1}\end{aligned}$$

We find the maxima to be:

$$N_{1M} = \frac{K_1 - \alpha_{12}N_2}{2}$$

Which only makes sense if $N_2 < \frac{K_1}{\alpha_{12}}$, otherwise the effective carrying capacity will be negative, which we cannot have. We can then find our MSY:

$$H_1(N_{1M}) = \frac{r_1}{4K_1} (K_1 - \alpha_{12}N_2)^2 = H_{1MSY}$$

When we set N_2 to 0, we get an MSY of the same form as in the previous section, confirming that the model gives a good estimate. We can do the same work to find our MSY for species 2

B Acknowledgements

I want to thank the members of my group, Sophie de Groot and Ida Olsson, for helping improve my model by providing feedback and helping me better understand the concept of competing species, given that I have not done Mathematical Biology before. The regular meetings deepened my understanding of the topic, allowing me to produce the model to the best of my ability. I would also like to thank the lecturer Andy Archer for his guidance in modelling, his constructive criticism, and for keeping us on the right track.

References

- [1] M. Stokes, “Population ecology at work: Managing game populations — learn science at scitable,” 2012.
- [2] “15.5: Quantifying competition using the lotka-volterra model,” 03 2022.
- [3] J. D. Murray, *Mathematical Biology*. Springer Berlin Heidelberg, 1993.
- [4] “M_{sy} -maximum sustainable yield,” 2012.
- [5] W. Contributors, “Maximum sustainable yield,” 01 2019.